Data Visualization Analysis

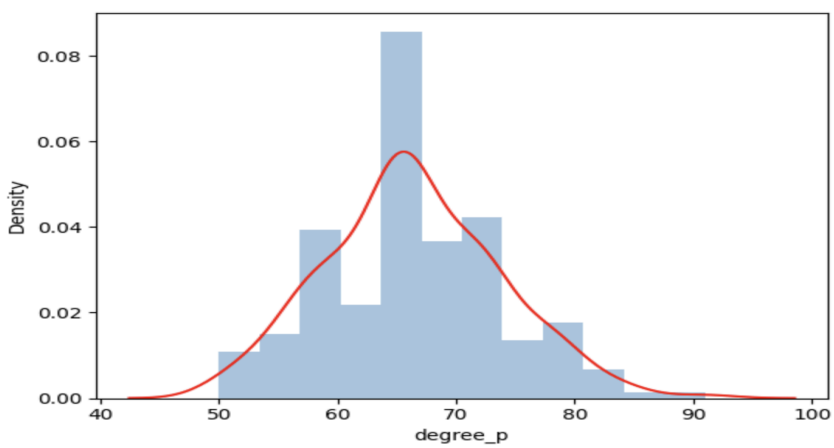
Placement Dataset

This dataset contains 215 rows and 15 columns.

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	workex	etest_p	specialisation	mba_p	status	salary
0	1	M	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech	No	55.0	Mkt&HR	58.80	Placed	270000.0
1	2	M	79.33	Central	78.33	Others	Science	77.48	Sci&Tech	Yes	86.5	Mkt&Fin	66.28	Placed	200000.0
2	3	M	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt	No	75.0	Mkt&Fin	57.80	Placed	250000.0
3	4	M	56.00	Central	52.00	Central	Science	52.00	Sci&Tech	No	66.0	Mkt&HR	59.43	Not Placed	NaN
4	5	M	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt	No	96.8	Mkt&Fin	55.50	Placed	425000.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
210	211	M	80.60	Others	82.00	Others	Commerce	77.60	Comm&Mgmt	No	91.0	Mkt&Fin	74.49	Placed	400000.0
211	212	M	58.00	Others	60.00	Others	Science	72.00	Sci&Tech	No	74.0	Mkt&Fin	53.62	Placed	275000.0
212	213	M	67.00	Others	67.00	Others	Commerce	73.00	Comm&Mgmt	Yes	59.0	Mkt&Fin	69.72	Placed	295000.0
213	214	F	74.00	Others	66.00	Others	Commerce	58.00	Comm&Mgmt	No	70.0	Mkt&HR	60.23	Placed	204000.0
214	215	M	62.00	Central	58.00	Others	Science	53.00	Comm&Mgmt	No	89.0	Mkt&HR	60.22	Not Placed	NaN

215 rows × 15 columns

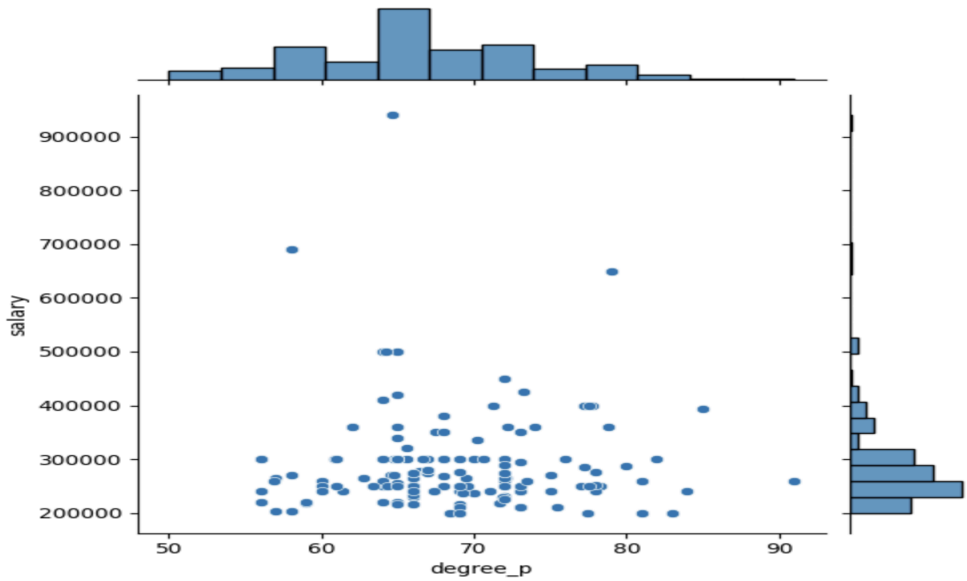
Univariate Analysis - Histogram



- I performed Univariate Analysis using Histogram plot for **Degree** Percentage (degree_p).
- The blue bars represent the **frequency** of data points falling into specific ranges from 50 to 90. The height of the bar indicates how many observations exist in that interval.
- **X-axis (degree_p):** This represents the value of the variable. Most of the data falls between **50** and **90**.
- **Y-axis (Density):** Density ensures the total area under the red curve equals **1**. This is useful for comparing distributions of different sizes.
- **Central Tendency (The Peak):** The highest peak (the mode) occurs around **65–70**. This tells you that the most common score in this dataset is in the low 60s.

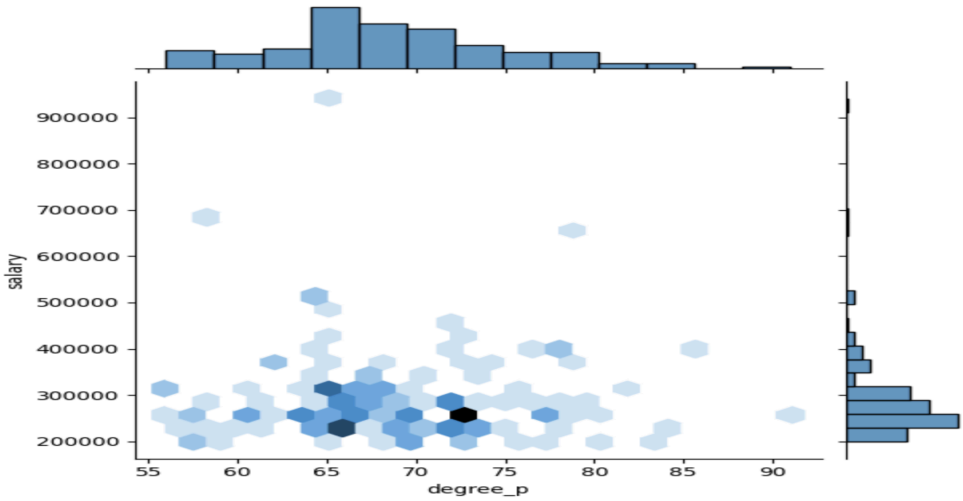
Bivariate Analysis

Joint plot



- The joint plot reveals a weak correlation between academic performance and salary.
- The histogram in **X-axis** shows the distribution of **degree_p**. The peak is around **65% to 75%**, indicating that most students graduated with those marks.
- The Y-axis shows the distribution of **salary**, most salaries are between **200,000 and 300,000**.

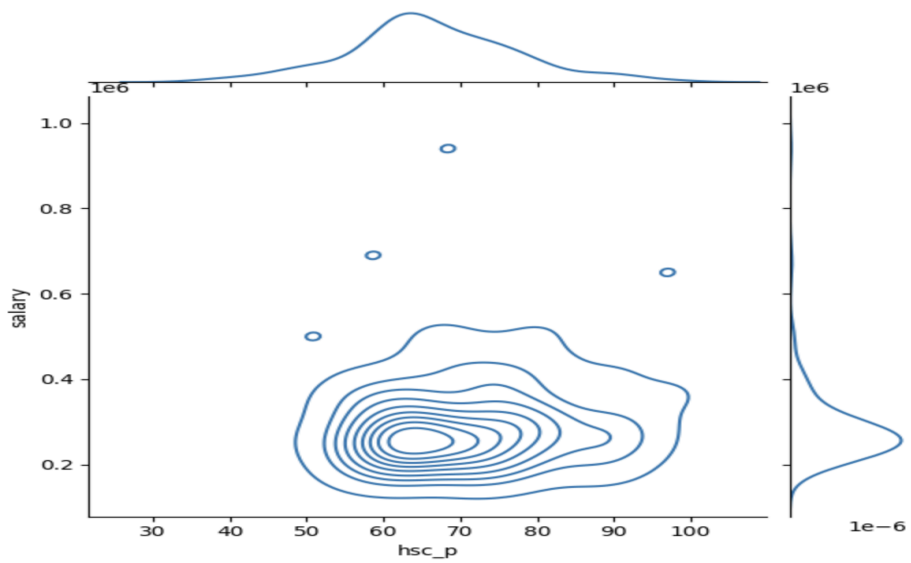
Hexbin Joint plot:



- It's very similar to the scatter joint plot, it uses **hexagonal bins** instead of dots.
- The darker the hexagon represents, the more data points are packed inside it.

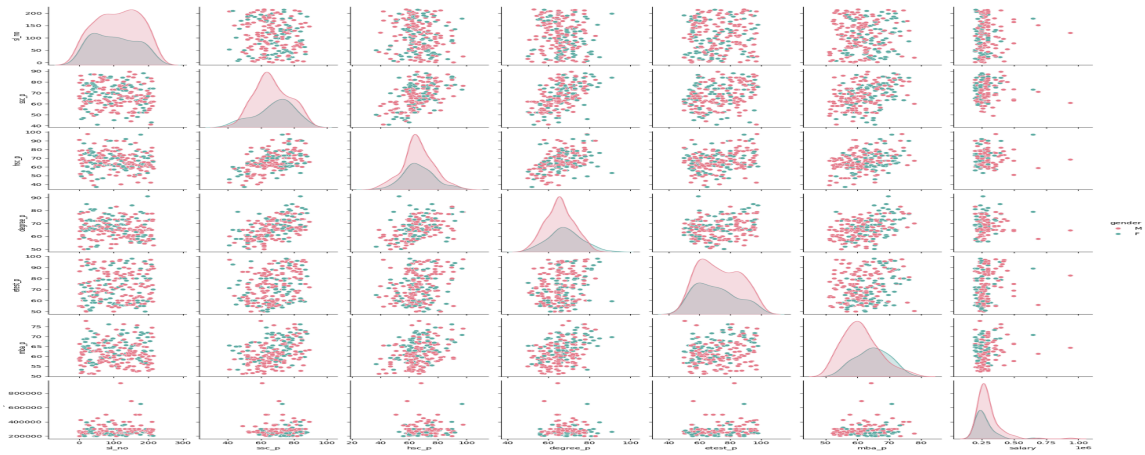
- Notice that one almost-black hexagon around **`degree_p = 73`** and **`salary = 250,000`**. The most common combination of degree score and salary in the entire dataset.

Kernel Density Estimation (KDE)



- In this joint plot I changed the ‘kind’ to ‘kde’. Instead of points we can see the concentric rings.
- The smallest, innermost circles represent the areas of **highest density**.
- As the rings spread out, the density of data points decreases.
- The KDE joint plot illustrates a high density of candidates scoring between 60% and 70% in their HSC, with a corresponding salary concentration in the 200,000–300,000 range.

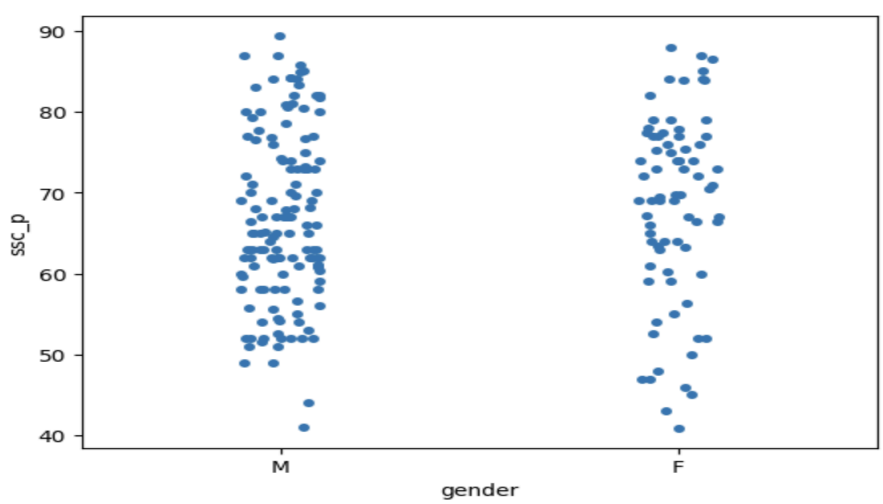
Pair Plot



- This pair plot transforms the scatter points into a "heatmap" of density.
- This reveals a primary density cluster at the intersection of a 65% HSC score and a 250,000 salary.
- The concentric rings around the salary axis indicate low variance for most candidates, while the upward vertical stretching of the outer contours suggests that higher salaries are statistically rare and not strongly tied to a specific HSC percentage.

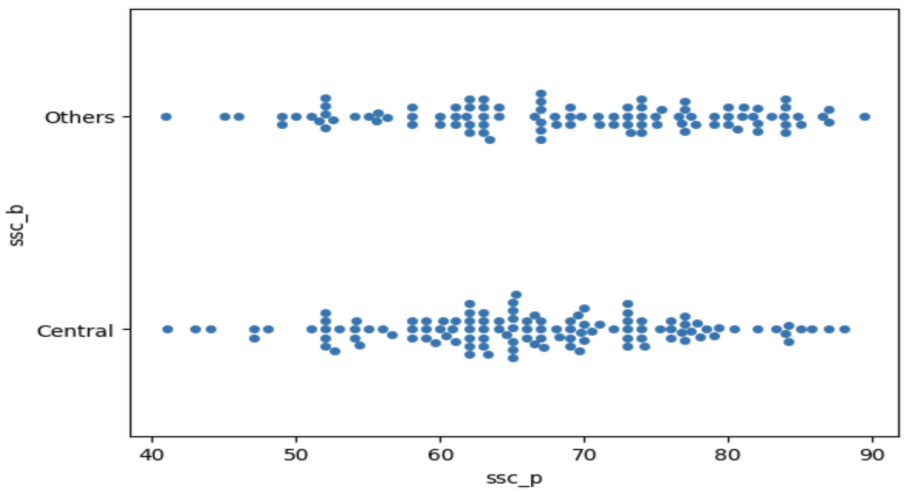
Categorical plot

Strip Plot:



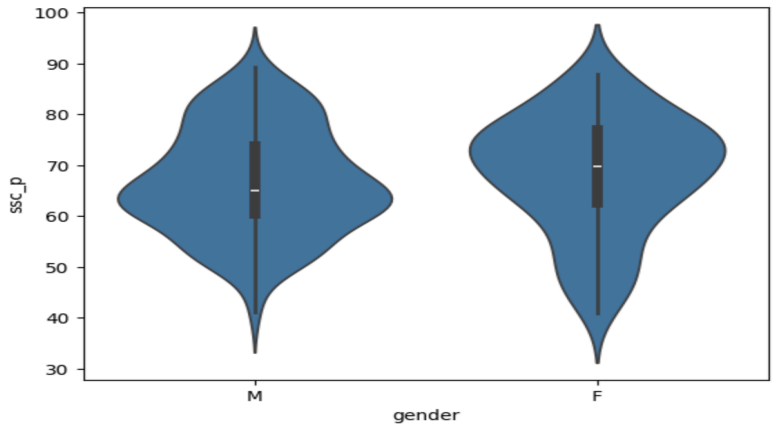
- This explains the distribution of data points of ssc_p between male and female students.
- This graph shows the number of male and female students. Most male students scored between 50 - 85% with few students scoring between 40 - 50%. One student scored 90%.
- The most female students scored between 50 - 80% and some students have average performance scored between 40 - 50%. The top performers scored between 80 - 89%.

Swarm plot:



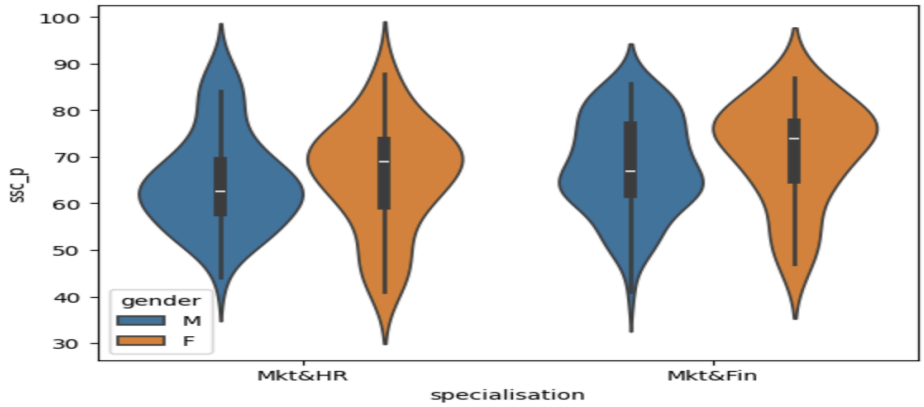
- The Swarm plot is a type of categorical strip plot which helps to visualize the distribution of values across different categories.
- **X-axis (ssc_p)** represents a numerical variable score ranging from approximately 40 to 90. **Y-axis (ssc_b)**: This represents a categorical variable with two distinct groups in ssc_b (Board) **Central** and **Others**.
- **Distribution**: Both categories ("Central" and "Others") show a fairly similar spread of scores across the x-axis.
- **Density**: The high concentration of data points in the **60 to 80** range for both groups.

Violin Plot (1)



- This violin plot compares the distribution of a variable called **ssc_p** percentage score between two groups: **Male (M)** and **Female (F)**.
- The white markers (Median) for both M and F are positioned at roughly **65 to 70**. This suggests that the score is quite similar for both genders.
- **Males (M):** The violin is widest around the **60–65** range, suggesting a large cluster of scores just below the median.
- **Females (F):** The violin is wider at a slightly higher point, around **70–75**, and has a sharper taper at the bottom.

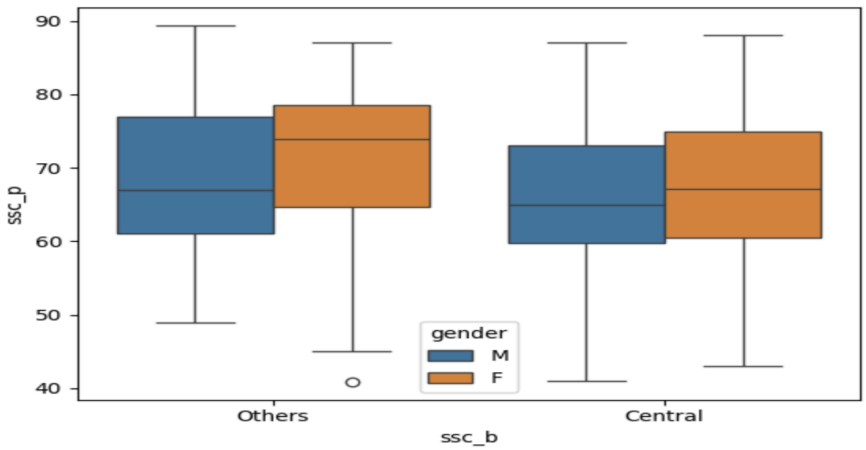
Violin Plot (2)



- This violin plot describes the relationship between ssc_p percentage, specialization (Marketing & HR vs. Marketing & Finance), and gender.

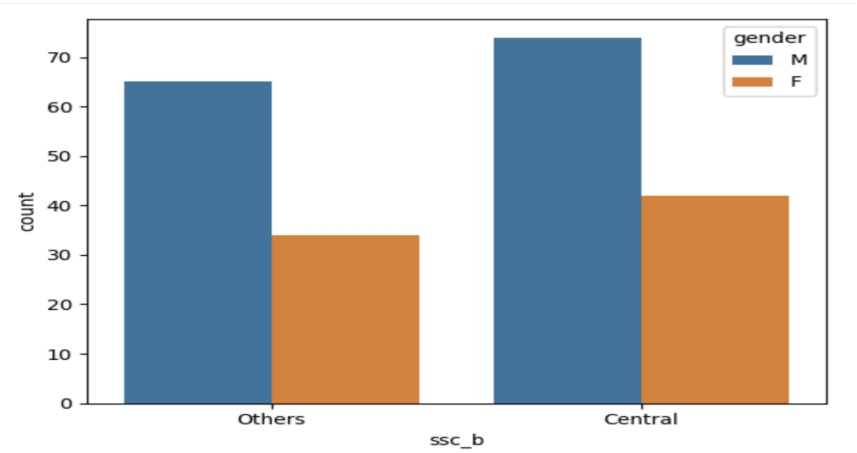
- Looking at the overall height and the "bulge" of the violins, students who specialized in Marketing & Finance generally had higher ssc scores than those in Marketing & HR.
- The orange violin for **Mkt&Fin** is shifted noticeably higher than the others. Its median (the white dot) is the highest on the chart, sitting near **75**, and a large portion of this group scored between **70 and 85**.
- For the **Mkt&HR** group, both males and females have very similar medians (around **62–68**). However, the female distribution is more "stretched," showing a wider variety of scores, while the males are more tightly clustered.

Box plot



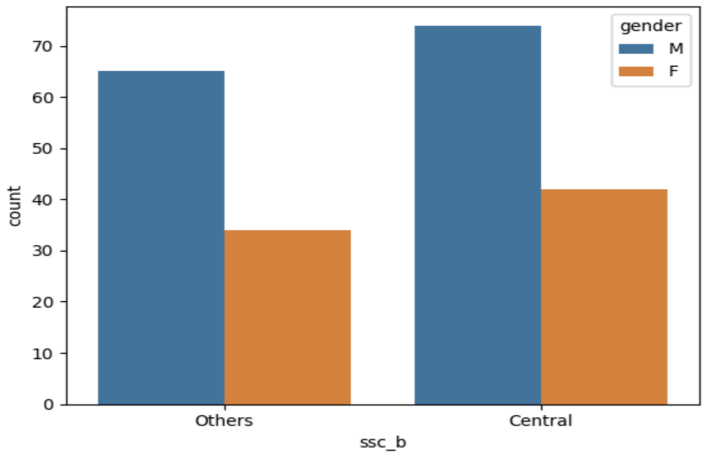
- This plot analyzes the ssc scores by board and gender.
- The male students from other boards scored between 62 to 76 %, the highest score was 86%.
- The female students from other boards scored between 63 to 78 %, the highest score being 89%.
- Of the students who chose Central board, the male students scored between 60 - 75% and the female students scored between 61 to 77%.
- The middle line represents the Median value among male and female students across the board.

Bar Plot



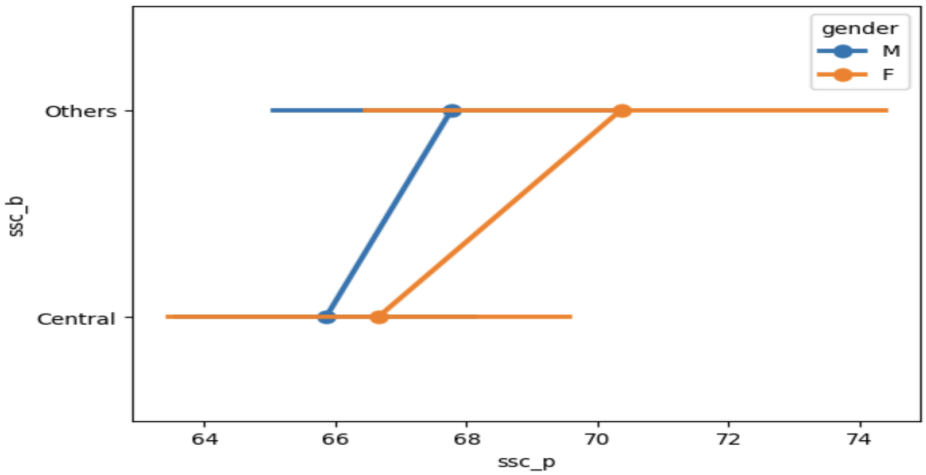
- This image analyzes the distribution of the SSC board among others and the Central board.
- Compare to other, the Central board students got high score in their Secondary

Count Plot



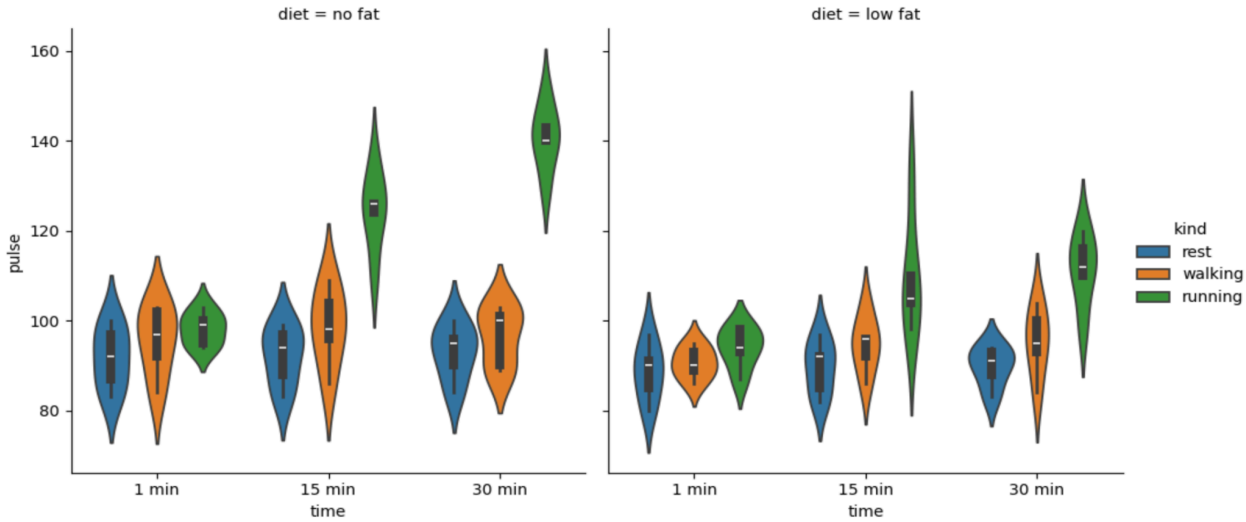
- Using a count plot we can analyze the frequency of students choosing different boards with respect to their gender.
- In other boards, the number of male students is 65 and the female students are 35.
- In Central boards, the number of male students is 75 and the female students are 45, which is higher than other boards.

Point Plot



- The Point plot is used to show the estimate of central tendency (Mean) and the uncertainty around that estimate for categorical data.
- **X-axis** represents a numerical score or percentage **Secondary School Certificate percentage**.
- **Y-axis** is a categorical variable with two different groups: **Central** and **Others**.
- The **Points** represent the **estimate of central tendency**, usually the **mean** (average). For example, females in the "Others" category have a higher average score (71%) than males in the same category (67%).
- **The Horizontal Lines (Error Bars)** indicate the **uncertainty** around that average, typically representing a **95% confidence interval**.
- **The Connecting Lines** are mainly used to highlight the **slopes** between categories. This helps you quickly see if the "gap" between groups changes as you move from "Central" to "Others."

Cat Plot



- This visualization is a violin plot created using Seaborn's Catplot which shows the distribution of pulse rates across different exercise, durations, and dietary conditions.
- The **X-Axis** represents **Time** Duration of the activity: 1 min, 15 min, 30 min. The **Y-Axis** represents the **Pulse** (Heart rate).
- **Color** displays the Activity type (Resting, Walking, Running) and the **Facets** (Columns) represents Diet types ("no fat" vs. "low fat").
- **Running (Green)** shows the most significant increase in pulse across all time intervals. It also has the widest variance, especially at 15 and 30 minutes, indicating that different individuals respond very differently to sustained running.
- **Walking (Orange) & Resting (Blue)** remain relatively stable. The pulse for walking is slightly higher than resting, but neither shows the dramatic upward trend seen in the running group as time increases.
- As time increases from 1 min to 30 min, the "Running" pulse distribution shifts upward and becomes more spread out.
- For "Rest" and "Walking," time has a negligible effect on the average pulse, suggesting these activities reach a steady state almost immediately.
- For the **"No Fat"** group the pulse rates for runners at 30 minutes are notably higher, peaking near **140–160 bpm**.
- The **"Low Fat"** Group - The runners at 30 minutes show a lower peak pulse compared to the "no fat" group, centering more around **110–120 bpm**.